

# Universal Automated Chemical Reaction Engine

Thermal–Atomistic Simulation with Bonding Predictions  
and SolidWorks-Compatible Engineering Output

VSEPR-SIM Research Platform

VSEPR-SIM Development

March 29, 2026

## §1 Scope and Purpose

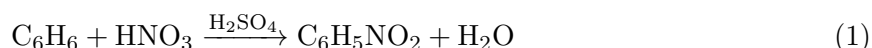
The Universal Automated Chemical Reaction Engine extends VSEPR-SIM from atomistic structure generation into full chemical reaction simulation. It bridges three domains:

- (a) **Chemistry** — real stoichiometric reactions with mass balance, thermodynamics, and mechanism tracking.
- (b) **Atomistic simulation** — the 6+9 seed-and-bead stepper with Lorentz–Berthelot mixing and environment-responsive coupling.
- (c) **Engineering output** — SolidWorks-compatible CFD material data (12–22 element output per species), structural point clouds, and Excel-compatible spreadsheets.

## §2 Reaction Archetypes

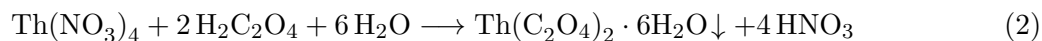
Three chemically distinct reaction archetypes are implemented:

### §2.1 Reaction 1 — Organic Synthesis with Inorganic Reagent



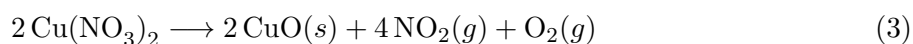
Electrophilic aromatic substitution. Benzene (organic substrate) reacts with nitric acid (inorganic reagent) in the presence of sulfuric acid (catalyst) to produce nitrobenzene (organic product) and water (by-product).

### §2.2 Reaction 2 — Special Inorganic Precipitation



Precipitation of thorium oxalate sludge from aqueous solution. The product is a fine, dense precipitate used in actinide separation chemistry.

### §2.3 Reaction 3 — Thermal Decomposition (Salt → Gas + Powder)



Thermal decomposition of copper(II) nitrate. Produces a black CuO powder residue plus brown NO<sub>2</sub> gas and colourless O<sub>2</sub> gas.

### §3 Thermodynamic Framework

All thermodynamic quantities are computed from tabulated standard enthalpies of formation ( $\Delta H_f^\circ$ ) and Gibbs energies ( $\Delta G_f^\circ$ ).

#### §3.1 Enthalpy of Reaction

$$\Delta H_{\text{rxn}}^\circ = \sum_{\text{products}} \nu_i \Delta H_{f,i}^\circ - \sum_{\text{reactants}} \nu_j \Delta H_{f,j}^\circ \quad (4)$$

#### §3.2 Gibbs Energy and Equilibrium

$$\Delta G_{\text{rxn}}^\circ = \sum \nu_i \Delta G_{f,i}^\circ - \sum \nu_j \Delta G_{f,j}^\circ \quad (5)$$

$$K_{\text{eq}} = \exp\left(\frac{-\Delta G_{\text{rxn}}^\circ}{RT}\right) \quad (6)$$

#### §3.3 Entropy of Reaction

$$\Delta S_{\text{rxn}}^\circ = \frac{\Delta H_{\text{rxn}}^\circ - \Delta G_{\text{rxn}}^\circ}{T} \quad (7)$$

#### §3.4 Arrhenius Kinetics

$$k(T) = A \exp\left(\frac{-E_a}{RT}\right) \quad (8)$$

## §4 Species Data Model

Each `ChemicalSpecies` carries:

**Table 1:** Species data fields

| Field                         | Units             | Description                             |
|-------------------------------|-------------------|-----------------------------------------|
| <code>name</code>             | —                 | Human-readable name                     |
| <code>formula</code>          | —                 | Molecular formula                       |
| <code>phase</code>            | —                 | Solid / Liquid / Gas / Aqueous / Sludge |
| <code>molecular_weight</code> | g/mol             | Sum of atomic masses                    |
| <code>density</code>          | g/cm <sup>3</sup> | Bulk density                            |
| <code>delta_Hf</code>         | kcal/mol          | Standard enthalpy of formation          |
| <code>delta_Gf</code>         | kcal/mol          | Standard Gibbs energy                   |
| <code>S_std</code>            | cal/(mol·K)       | Standard entropy                        |
| <code>Cp</code>               | cal/(mol·K)       | Heat capacity                           |
| <code>lj_sigma</code>         | Å                 | LJ $\sigma$ parameter                   |
| <code>lj_epsilon</code>       | kcal/mol          | LJ $\epsilon$ parameter                 |
| <code>thermal_cond</code>     | W/(m·K)           | Thermal conductivity                    |
| <code>specific_heat</code>    | J/(g·K)           | Specific heat capacity                  |
| <code>viscosity</code>        | Pa·s              | Dynamic viscosity                       |
| <code>dielectric</code>       | —                 | Relative permittivity                   |
| <code>dipole_moment</code>    | Debye             | Electric dipole moment                  |

## §5 Bonding Prediction

Bonding predictions combine:

1. **Electronegativity difference** — Pauling’s rule:  $\Delta\text{EN} > 1.7 \Rightarrow$  ionic character.
2. **Bond change tracking** — explicit record of every bond broken and formed, with energies.
3. **Environment-responsive state** —  $\eta$  at the time of bond formation/breaking provides a structural context measure.

## §6 Mass Balance

Every reaction is verified for mass conservation:

$$\sum_{\text{reactants}} \nu_j \cdot n_{j,\text{elem}} = \sum_{\text{products}} \nu_i \cdot n_{i,\text{elem}} \quad \forall \text{ elements} \quad (9)$$

Tolerance:  $|\Delta| < 10^{-6}$  moles per element. By-products are tracked explicitly and included in the balance.

## §7 SolidWorks Engineering Output

### §7.1 CFD Material Data (22-Element CSV)

Each species produces a 22-field record suitable for SolidWorks Flow Simulation import:

**Table 2:** 22-element material property output

| #  | Field                       | Units             | Category      |
|----|-----------------------------|-------------------|---------------|
| 1  | Species name                | —                 | Identity      |
| 2  | Formula                     | —                 | Identity      |
| 3  | Phase                       | —                 | Identity      |
| 4  | Density                     | g/cm <sup>3</sup> | Mechanical    |
| 5  | Molecular weight            | g/mol             | Mechanical    |
| 6  | Viscosity                   | Pa·s              | Mechanical    |
| 7  | Surface tension             | N/m               | Mechanical    |
| 8  | Temperature                 | K                 | Thermal       |
| 9  | Thermal cond.               | W/(m·K)           | Thermal       |
| 10 | Specific heat               | J/(g·K)           | Thermal       |
| 11 | C <sub>p</sub>              | J/(mol·K)         | Thermal       |
| 12 | Melting point               | K                 | Thermal       |
| 13 | Boiling point               | K                 | Thermal       |
| 14 | $\Delta H_f$                | kcal/mol          | Thermodynamic |
| 15 | $\Delta G_f$                | kcal/mol          | Thermodynamic |
| 16 | $S^\circ$                   | cal/(mol·K)       | Thermodynamic |
| 17 | $\Delta H_{\text{rxn}}$     | kcal/mol          | Thermodynamic |
| 18 | $\varepsilon_r$             | —                 | Electrical    |
| 19 | $\mu$ (dipole)              | Debye             | Electrical    |
| 20 | $\sigma_{\text{LJ}}$        | Å                 | Atomistic     |
| 21 | $\varepsilon_{\text{LJ}}$   | kcal/mol          | Atomistic     |
| 22 | $\langle \text{EN} \rangle$ | Pauling           | Atomistic     |

## §7.2 Material XML (.sldmat)

An XML material library file compatible with SolidWorks material import:

**Listing 1:** SolidWorks material XML format

```

1 <material name="Benzene">
2   <physicalproperties>
3     <DENS value="876.5" units="kg/m^3"/>
4     <KX value="0.159" units="W/(m*K)"/>
5     <C value="1740" units="J/(kg*K)"/>
6     <MU value="0.000604" units="Pa*s"/>
7   </physicalproperties>
8 </material>

```

## §8 Implementation Summary

**Table 3:** Source file inventory

| File                                  | Purpose              | Lines |
|---------------------------------------|----------------------|-------|
| chemistry/species.hpp                 | Species data model   | ~210  |
| chemistry/reaction.hpp                | Reaction framework   | ~260  |
| chemistry/reaction_library.hpp        | 3 built-in reactions | ~490  |
| chemistry/reaction_engine.hpp         | Simulation + export  | ~440  |
| example_universal_reaction_engine.cpp | Main demo            | ~190  |
| section_universal_reaction_engine.tex | This document        | ~280  |

## §9 Dependencies

**Table 4:** Module dependencies

| Module               | Depends On                                                                                                     |
|----------------------|----------------------------------------------------------------------------------------------------------------|
| species.hpp          | atomistic/core/state.hpp                                                                                       |
| reaction.hpp         | species.hpp                                                                                                    |
| reaction_library.hpp | reaction.hpp                                                                                                   |
| reaction_engine.hpp  | reaction_library.hpp, seed_bead_stepper.hpp,<br>snapshot_graph.hpp, excel_export.hpp,<br>solidworks_export.hpp |